

Systemic Arrestability: A Temporal Criterion for Machine Safety

Operator protection when machine speed exceeds human reaction

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ABSTRACT

Machine safety law protects the operator in part through instruments — the stop function, the emergency stop — whose operation presupposes that a human being can perceive a developing hazard, evaluate it, and act before harm occurs. That presupposition can be tested. It amounts to a comparison between the interval a hazard allows and the window that stopping it requires, and the comparison is decisive: where it fails, no instrument placed in the operator's hand can perform the protective function the law requires.

The available interval runs from the onset of a hazardous action to the point beyond which its effect can no longer be prevented. Against it stands the window human intervention requires: the minimum credible time to perceive the hazard, decide, and act, for the configuration in which the exposed person stands. Where that window is longer than the interval available, the chain of human intervention cannot complete. A protective architecture whose last layer depends on that chain then provides no protection at all — not reduced protection, but none. The capacity to halt the hazardous action must instead reside in a mechanism whose own window fits the interval available, and that does not share a failure path with what produced the hazard. Where no such mechanism exists, that configuration cannot be shown to satisfy the duty of safe design.

The comparison is established first from the mechanics of the hazard and the physiology of response, without reference to any legal system. European Union machinery law, Swiss product-safety law and United States occupational safety regulation are then shown already to require the result the comparison produces, while none of them states the comparison itself. The criterion of systemic arrestability names that comparison and puts it in a form that can be demonstrated for a particular configuration. It adds no obligation to existing law. It states, in terms that admit proof, what the duty of safe design has always required.

1. Introduction

A machine must be designed and operated so that hazardous action can be stopped before it causes harm. In the ordinary case, machine-safety law places part of that capacity in the operator's hands through the stop function, the emergency stop and related controls. The principle is not aspirational: it is reflected in the stop-function and emergency-stop requirements of Directive 2006/42/EC, in the product-safety duties of the Swiss Federal Act on Product Safety, and in the machine-guarding requirements of 29 CFR § 1910.212. The same protective architecture recurs well beyond these systems, carried by international standards — ISO 12100 above all — that give machine-safety duties their operational content and are adopted as national standards in jurisdictions that share little else.

Two of those instruments are placed in a human hand. An emergency stop presupposes a hand that reaches it while the danger is still impending. A stop function presupposes an operator in whom the decision to use it can form. What they share is not a technology but a premise: that between the onset of a hazardous action and the occurrence of harm there is time for a human being to act. Physical guarding is the case in which the law has already dispensed with that premise, and Section 3.4 returns to it.

That premise can be tested, because the interval and the window can both be measured. This paper states the comparison, follows it to its end, and shows what it decides.

The order of the argument is the reverse of the usual one. Section 2 works without reference to any legal system. The available interval is fixed by the mechanics of the hazard, and the window a person needs by the physiology of response, and what follows from them follows wherever machines are built. Section 3 then examines European, Swiss and United States machine-safety law. Each already imposes the result the comparison produces; none states

the comparison itself. Section 4 shows that the same silence runs through the technical standards on which those systems converge, so that the lacuna is common to regimes that agree on little else. Section 5 gives the result a name — systemic arrestability — and shows that industrial practice has satisfied it for decades. Section 6 states what the criterion does in operation.

It also states where work remains. The response time of a halting mechanism is standardised in detail, and so is the interval a hazard allows. The window human intervention requires has never been consolidated as a reference value at all.

Article 3(1) LSPro permits a product to be placed on the market only if it exposes users and third parties to no danger, or only to minimal danger. EHSR 1.1.2 requires risks to be eliminated by design before protective measures are taken and before information is given to the user. Section 1910.212(a)(1) requires that some method of guarding actually protect the operator. Each of these is in force. The comparison only makes it possible to say, of a given configuration, whether what they require was done or only documented.

The analysis covers civil and dual-use machinery only. The three frameworks it relies on do not reach military systems. The European machinery regime expressly excludes machinery designed and built for military or police purposes. Swiss product-safety law governs products placed on the market, and American occupational safety law governs civilian workplaces. Neither covers military operations.^{[1][2]} The protection of third parties is not argued separately. Section 6.3 shows why it needs no separate argument, and on what condition.

2. The comparison and what follows from it

The argument turns on one interval and the windows that must fit inside it. A configuration can fail in two ways, and the two are treated separately below. The interval a hazard allows may be shorter than the window human intervention requires, which is the case examined in 2.3. Or there may be no usable interval at all, which is the case examined in 2.6. In the first the capacity to halt must move into a mechanism. In the second no mechanism reaches it.

Value	What it is	Fixed by
Available interval	From the onset of a hazardous action to the point beyond which its effect can no longer be prevented	The mechanics of the hazard
Human response time	The minimum credible response time for the configuration in which the exposed person stands	Human physiology, within the configuration's constraints
Mechanism response time	The time the halting mechanism takes to act	The design of that mechanism

2.1 The interval the hazard allows

An effect is operationally significant when the causal sequence producing it has passed the point beyond which no further intervention — of any kind, by any actor, at any speed — can prevent it. That point is fixed by the hazard itself: the moment a moving mass can no longer be arrested, a release can no longer be recalled, an energy transfer can no longer be interrupted. It is a property of the physical process, not of any observer's capacity to respond to it. Irreversibility here is not a typical indicator of operational significance; it is the definition. That the effect carries injury is a separate and confirmatory matter, and it does no defining work.

The question is never whether a machine is fast. Many machines perform fast motions inside guarded spaces, and the guarding is what makes their speed legally irrelevant: no one is exposed, so there is nothing to measure. The question is whether the machine can produce operationally significant effects, where persons are exposed, faster than those persons can respond.

2.2 The window human intervention requires

Stopping a machine is not a single act. It is a chain of events that has to run from the hazard to the control and back to the machine, and every one of them takes time. The hazard must produce a perceptible signal. The signal must be perceived. The perception must be evaluated and a decision formed. The decision must be executed as a physical act on a control. The control must take effect on the machine. The times compound.

The empirical literature gives orders of magnitude rather than fixed constants, and nothing more is needed here. Simple reaction to an expected stimulus is short. Recognition and choice among alternatives lengthen it. Realistic operational conditions lengthen it further: an unexpected hazard, an ambiguous signal, a control at arm's length rather than under the finger. What the literature establishes is that the chain has a floor, and that the floor is not a matter of effort.^[3]

No standard establishes that floor. Section 6.2 returns to this.

A person has a minimum reaction time that nothing can lower. It is physiology. The way a machine is built can only make it worse: a control out of reach, an unclear signal, a hazard that becomes visible late. The physiological minimum is therefore the best case, and the real figure for a given configuration is always above it.

That figure can be established before the machine is placed on the market. It depends on the machine and on the operating position, both of which are known at that stage. The time human intervention requires is therefore not an imponderable. It is a parameter, and design can be measured against it.

2.3 Where the interval is shorter, protection is absent rather than reduced

Where the interval available is shorter than the window that chain requires, it cannot complete before the effect occurs. The most capable operator conceivable stands in exactly the same position. The emergency stop exists; the hand cannot reach it. The stop function exists; the decision to use it cannot form.

Conformity is assessed by checking that the required devices are present and function. It is not assessed by asking whether they can be used in time. So the devices are installed, conformity is declared, the file is complete — and none of it registers what has just been described. The machine is formally compliant and functionally unguarded.

The point is worth stating once in general form, because it explains why this kind of failure is silent. A rule that protects a person through an operation presupposes the duration that operation takes. The presupposition is not a further requirement the rule imposes; it is the condition of the rule doing anything at all, since an operation carried out in no time is not a hurried operation but an absent one. Where a system is designed so that no such duration exists, the rule remains in force and ceases to operate: it is never contradicted, only emptied. Which operation is protected, and on whom the duty falls, is a matter for each legal regime; nothing is extended here beyond the machine-safety duties examined in this paper, where the operation is the stopping of a hazardous action and the duty is the manufacturer's.

2.4 The halting capacity must lie in a mechanism that acts within the interval

If the hazardous action is to be halted at all in such a configuration, the capacity to halt it must sit in a mechanism whose own window fits the interval the hazard allows. The mechanism may be physical, logical or architectural. Which of them is used is a matter for the designer: any arrangement that acts within the interval satisfies this, and none that does not.

2.5 The mechanism's response time is defined only where it is independent of what caused the hazard

A stopping mechanism is only as good as what it depends on. Take a machine driven by a controller. The controller fails and leaves the drive running. If the stop command is sent to that same controller, it reaches a component that has already failed. The fault that created the hazard has also removed the remedy.

The stopping path must therefore be separate from what can fail. A dedicated relay cutting power upstream is separate; a stop routine inside the failed controller is not. Where it is not separate, a response time cannot be

claimed for the mechanism at all: the figure measured in normal operation says nothing about the case in which the mechanism is needed.

Separation is easy to arrange when the hazard comes from a component that can be cut off. It is harder when the hazardous action is decided by a model. A stopping mechanism that reads the same sensors and runs on the same network is not separate from the decision it is meant to catch: an error in that decision reaches both, and the mechanism confirms the action instead of stopping it.

What remains separate in such a system is whatever acts on a physical quantity instead of on an interpretation of the scene. A power limit, a dedicated proximity sensor, a physical constraint on the working space: none of these depends on what the model concluded. Where a configuration offers nothing of this kind, and the only thing that could stop it shares the decision path, the case is the one set out in 2.6.

2.6 Where the interval admits no mechanism

For some hazards no interval opens at all. The point beyond which the effect can no longer be prevented coincides with the onset of the hazardous action. For others it opens too late to be used: the hazard becomes detectable only after that point, so a mechanism of any speed would be acting on a signal that arrives once the outcome is settled. Neither case is a short interval that a faster mechanism could act within.

No mechanism acts after the closing point of the interval. That point was defined in 2.1 as the point beyond which no intervention of any kind, by any actor, at any speed, prevents the effect.

Section 2.7 sets out what the demonstration contains. Where the interval admits no mechanism, that demonstration cannot be produced, and the configuration is not shown to satisfy the duty of safe design.

What is assessed is not the machine but the way it is installed. The interval and the window both depend on an operating position, a separation distance and an exposure condition. The same machine may pass inside an enclosed cell and fail in a shared workspace: the machine has not changed, the installation has. So when a configuration is not shown to be safely designed, that is a statement about that installation, not about the machine.

One distinction should be held firmly, because it is the point at which the comparison can be dissolved. A mechanism that is too slow, or insufficiently independent, is a design shortcoming, and engineering or research may close it; if it is closed, the demonstration can be produced and the configuration passes. An interval that never opens, or that opens only after the outcome is settled, is not a limit of the art but a property of the hazard, and no work closes it. If the two are allowed to blur, any configuration may be defended indefinitely on the ground that a faster mechanism might yet appear. Nothing in Section 2.7 requires the distinction to be adjudicated in advance: it matters to whoever does the research, not to whoever performs the assessment.

2.7 The operative form is an evidentiary burden

The distinction just drawn does not need to be adjudicated, because the chain does not ask anyone to prove a negative. The burden runs the other way, and it is always positive.

The burden is anterior, and it falls on whoever places the machine on the market. It is discharged before the machine reaches the place where it can do harm, not after an accident has raised the question — which is what makes the comparison a condition of placing on the market rather than a standard of liability. For the configuration as it will be used, that person demonstrates:

1. the interval, established from the mechanics of the hazardous action;
2. the human response time, established for the operating position and the exposure conditions;
3. either that the interval is longer than the window human intervention requires, so that human intervention remains a justifiable part of the safety case; or
4. that a halting mechanism acts within the interval, together with the basis on which its response time is claimed, including the absence of a shared failure path with the source of the hazard.

Where this demonstration is not produced, the configuration is not shown to be safely designed. Nothing turns on whether the cause is a temporary limit of the art or a permanent property of the hazard: that distinction interests

whoever does the research, not whoever performs the assessment. Where the demonstration is produced later, on the strength of a mechanism that did not previously exist, the configuration passes. The criterion compares an interval with the windows that must fit inside it, and the values are those the state of the art can deliver at the moment of assessment.

This is also what makes Section 2 usable rather than merely valid. The manufacturer has a calculation to perform, the assessor a demonstration to inspect, and the adjudicator a fact to find.

3. Existing law already imposes the result

3.1 European Union: the hierarchy and the devices

The Machinery Directive operates through Essential Health and Safety Requirements that manufacturers must satisfy before placing machinery on the market. EHSR 1.1.2 establishes safety integration: risks are to be eliminated or reduced by design as the first and preferred measure, protective measures second, and information to users only as a residual instrument. The hierarchy already prefers solutions that do not depend on human conduct.

EHSR 1.2.4.1 requires a control device by which the machinery can be brought safely to a complete stop. EHSR 1.2.4.3 requires one or more emergency stop devices to enable actual or impending danger to be averted. Both provisions put the protection in a person's hands. A stop control has to be operated. An emergency stop has to be reached, by someone who has recognised the danger while it is still impending rather than consummated.

EHSR 1.2.2 requires control devices to be positioned so that they can be operated without hesitation, without loss of time and without ambiguity. EHSR 1.2.4.3 requires the emergency stop to halt the hazardous process as quickly as possible. The Commission's guidance gives the reason: in an emergency, a split-second reaction may be crucial.^[4]

In *CSF v Commission* the General Court held that a specific requirement of Annex I must be read in the light of the General Principles and of the safety-integration principles of EHSR 1.1.2. It held that the Directive does not merely oblige manufacturers to warn their customers of risks, but requires them to eliminate or reduce those risks as far as possible at the design and construction stage. And it held that once a risk is ascertained, prior notice of it to the user is in itself irrelevant, given the hierarchy of prevention and information requirements.^[5]

3.2 European Union: the correction requirement of Regulation (EU) 2023/1230

Regulation (EU) 2023/1230, applicable from 20 January 2027 in place of the Directive, addresses control systems of machinery with fully or partially self-evolving behaviour or logic designed to operate with varying levels of autonomy. For such systems it must be possible at all times to correct the machinery or related product in order to maintain its inherent safety. Such systems must not cause the machinery to act beyond its defined task and movement space. And the manufacturer's risk assessment extends to hazards arising as an intended evolution of the self-evolving behaviour.^[6]

The requirement speaks of *maintaining* inherent safety, not of adding to it. It sits in Annex III among the design requirements for control systems, not among operator instructions. And it attaches to a category: self-evolving behaviour as such.

Autonomy aggravates the temporal problem. A system that initiates hazardous action without a prior human command delays the perceptible signal, and the action arrives unanticipated: the operator is not waiting for it, and an unexpected event lengthens the response beyond an expected one. One that varies its behaviour lengthens evaluation, because the operator cannot rely on a learned pattern. Both effects widen the gap between the interval available and the window a person needs. But a non-autonomous machine producing irreversible effects in milliseconds presents the identical problem, and an autonomous machine slow enough to be interrupted presents none of it. The Regulation answers the problem for one class of machines. The problem arises wherever the comparison fails.

3.3 Switzerland: the product duty and its criminal backing

Article 3(1) LSPro permits a product to be placed on the market only if, used normally or under reasonably foreseeable conditions, it exposes the safety and health of users and third parties to no danger or only to minimal danger. Article 4 empowers the Federal Council to lay down essential health and safety requirements for specific product categories in sectoral legislation. That is the channel through which machine-specific requirements aligned with the European EHSRs enter Swiss law. The duty falls on the manufacturer, and it precedes use: safety must exist in the product as placed on the market.^[7]

Articles 16 and 17 LSPro attach criminal liability to violations of the product-safety duties. What the criminal provisions sanction is the placing on the market of a product that does not meet the safety requirements. If a product's safety case rests on operator intervention that the interval excludes, the product arguably never satisfied Article 3(1) at the moment it was placed on the market. The question for the manufacturer is not whether a stop device exists, but whether it can be used in time.

3.4 United States: guarding, and the refusal to rely on reaction

Section 1910.212(a)(1) requires that one or more methods of machine guarding be provided to protect the operator and other employees in the machine area from hazards such as those created by point of operation, ingoing nip points, rotating parts, flying chips and sparks.^[8]

The regulation does not prescribe a particular guard. Its own examples run from barrier guards to two-hand tripping devices to electronic safety devices. What it requires is that some method actually protect. A method that depends on the operator reacting faster than the hazard develops does not protect the operator: it leaves the operator to protect himself. One paragraph later the point is textual. Under § 1910.212(a)(3), point-of-operation guarding shall be so designed and constructed as to prevent the operator from having any part of his body in the danger zone during the operating cycle. Where reaction cannot be trusted, the regulation does not demand better reaction. It designs the operator out of the danger zone.

Work rules governing operator conduct are not a method of guarding contemplated by § 1910.212(a)(1). The adequacy of guarding turns on how the machine functions and is operated. And exposure is established wherever it is reasonably predictable that employees have been, are, or will be in the zone of danger.^[9]

Nip points, rotating parts and flying chips are fast hazards. American machine-guarding law was built around dangers that already exceeded human reaction time at the point of operation, and its answer was never to react faster. It was physical guarding that makes the interval humanly irrelevant.

3.5 What converges, and what is missing

Each of the three systems imposes the safety duty before and above the operator's own conduct: through safety integration, through conditions on market placement, through mandatory guarding. Each supplies stopping or guarding instruments whose ordinary operation presupposes human perception and reaction, and the European texts say so in terms. And in each, where human reaction cannot bridge the interval, the duty moves to design. EHSR 1.1.2 says it in its hierarchy. Section 1910.212 does it in its guarding logic. Regulation 2023/1230 states it for self-evolving machinery. The LSPro carries it in a duty that precedes use and is backed by criminal liability.

No provision in any of the three requires the window to be measured. None requires it to be set against the interval the hazard allows. And none states what follows when it does not fit. Without that measurement the manufacturer cannot show that the duty imposed by Article 3(1) LSPro and by the Essential Health and Safety Requirements has been discharged for the configuration in which the machine will be used.

This is the precise sense in which the problem is one of effectiveness rather than regulatory absence, and the distinction should be stated plainly. A regulatory gap exists where the law imposes no relevant duty and new legislation is needed to create one. That is not the problem here. The duty of safe design already exists. What is missing is an explicit criterion for determining, in a particular configuration, whether reliance on human intervention can actually discharge that duty. Without such a criterion a machine can satisfy every specific provision — devices fitted, conformity declared, documentation complete — and still leave the operator with no means of stopping it in the configuration in which it is used. The law supplies the obligation; the temporal

comparison supplies the proposition that connects the obligation to the facts. The first kind of gap calls for a legislator. The second calls for a criterion — which is why systemic arrestability adds no obligation and requires no enactment before it can be used.

4. The lacuna is common, because the standards are common

The same architecture recurs outside the three systems examined. Whether it does matters for one reason: if the silence identified in Section 3.5 is regional, it is a regional problem; if it runs through the instruments on which machine safety is built everywhere, that is where it sits.

The primary sources of the systems mentioned below are in Russian, Chinese and Hebrew. The treatment relies on published translations of binding texts and on the documented adoption of international standards. What is claimed is that the same protective structure recurs, not what any of these texts means in detail. The argument stands on Sections 2 and 3 and would stand without this section.

The Eurasian Economic Union regulates machinery through Technical Regulation TR CU 010/2011, binding since 2013 across its member states. Minimum safety requirements are imposed on the manufacturer across the lifecycle from design onward. Control systems must be equipped with emergency braking and emergency stop means wherever their use can reduce or prevent danger. The emergency stop device must remain in the stop position until deliberately reset, and its reset must not by itself restart the machine.^[10]

The People's Republic of China reaches the same architecture by a different route: the adoption of international standards as national ones. GB/T 15706 adopts ISO 12100 on safe design and risk reduction. GB/T 5226.1 adopts IEC 60204-1, which carries the emergency-stop requirements. GB/T 16855.1 adopts ISO 13849-1 on safety-related parts of control systems. Each is designated IDT — identical — in the national catalogue, adopted without substantive alteration.^[11]

Israel's framework is anchored in the Work Safety Ordinance (New Version) of 1970. It carries forward the British factories-legislation tradition from which American machine-guarding law also descends: the duty to securely fence dangerous parts of machinery. The mandatory standards declared under Israeli law are adoptions of the corresponding IEC and ISO standards.^[12] The same tradition carries an older instance of the point made throughout this paper: that the safety duty attaches to construction rather than to conduct.^[13]

The protective architecture examined in Section 3 is not a European inheritance. It is carried into legally and politically diverse systems by the same small family of international standards, chiefly ISO 12100, IEC 60204-1 and ISO 13849-1. Those standards define safety functions, performance levels and response times that do not depend on human reaction. They quantify the mechanism's side in considerable technical detail. What they do not establish is the window human intervention requires, nor any requirement to set it against the interval. Until that value is established, the comparison cannot be performed anywhere those standards apply.

As voluntary instruments these standards bind no court, and nothing here derives an obligation from them. That is part of why the criterion is proposed as a reading of binding duties rather than as a technical requirement. What they show is that the missing measurement is missing in the same place everywhere. A lacuna in ISO 12100 is a lacuna in Chinese practice, in European presumption of conformity, and in every national scheme that adopts it. That is also where correcting it would have the widest effect.

5. The criterion named

5.1 Systemic arrestability

A system is systemically arrestable when its capacity to halt a hazardous action does not depend on a human operator perceiving, evaluating and intervening within the interval that action allows.

The criterion states Section 2 as a single proposition, and carries no content Section 2 does not supply: where the interval is long enough for a person to act, human intervention remains a justifiable part of the safety case and the operator's devices retain their role; where it is not, the halting capacity must reside in a mechanism that acts within the interval and does not share a failure path with the source of the hazard; where no such mechanism can exist, the demonstration the duty requires cannot be produced for that configuration. What the criterion adds is not a requirement but a form — one in which the question can be asked of a particular machine and answered by a demonstration.

The criterion prescribes no engineering solution. Any mechanism satisfying the temporal and independence conditions satisfies it.

5.2 The criterion is already practised

Systemic arrestability is ordinary industrial practice. Its purest instances are the interlock and the scram.

A machine-tool interlock cuts power when a guard opens. A contact breaks a circuit; nobody has to notice anything. A reactor scram releases the control rods on a parameter excursion, and they fall under gravity, because no control-room crew can be relied on to react within the relevant interval. Neither mechanism asks the control system for permission, and neither waits for a person. That is what gives their response times a value in the conditions that matter.

These belong to a broader family of safety functions engineered to act below the threshold of human reaction, from anti-lock braking to airbag deployment. The family shows how routine human-independent response has become. The interlock and the scram remain the canonical cases because what they do is precisely arrest.

The criterion demands nothing the state of the art has not delivered routinely, across industries, for decades. Nor has it priced machine tools or power plants out of existence: interlocks and scram circuits are standard equipment. A criterion already satisfied by an interlock is not a burden invented for emerging technology.

5.3 A worked example: the collaborative robot

Consider a collaborative robot sharing an unfenced workspace with a human. The robot's arm can execute a movement capable of injurious contact.

At typical collaborative speeds and separation distances, the interval between the onset of such a movement and contact is on the order of a few hundred milliseconds. The window an unexpected movement requires of a human being is on the order of one second or more. The comparison fails.

Under a traditional conformity reading the installation may nonetheless present as compliant: an emergency stop is fitted and reachable, the instructions require the operator to remain attentive, the documentation is complete. Under the criterion, the comparison settles the question before the documentation is opened. Reliance on the operator is not a justifiable design choice for this configuration. The halting capacity must reside in the system: here, typically, a protective stop triggered by proximity or force sensing. Its sensing and stopping path does not depend on the controller executing the movement.

Well-designed collaborative installations are engineered exactly this way, through speed-and-separation monitoring and power-and-force limiting. The criterion demands no new engineering. It states the legal reason why the existing engineering is not optional.

For the well-designed machine the two readings give the same outcome. They differ for the badly designed one. Under the traditional reading, the formally compliant but functionally unguarded installation must be attacked indirectly, through the adequacy of its risk assessment. Under the criterion it fails directly, because the duty of safe design, read against the actual interval, was never discharged.

6. Operation, open parameters, scope

6.1 What the criterion does at three junctures

For the **manufacturer**, the comparison becomes a step of the risk assessment. The interval and the window are established for the configuration. Reliance on operator intervention becomes a design choice that must be justified by the comparison rather than assumed — at the design stage, where the duty of safe design places it, and not in the instructions supplied with the finished machine.

For the **conformity assessor**, it supplies the question: does the halting capacity fit inside the interval the hazardous action allows, independently of human reaction, and on what basis is that claimed?

For the **adjudicator**, it gives content to a duty whose breach is otherwise hard to articulate. A machine that depended on humanly impossible intervention did not satisfy the duty of safe design, however complete its documentation. The existing duty, read against the actual interval, was never discharged. This is the reading method the General Court applied in *CSF v Commission*: a specific requirement of Annex I interpreted in the light of the general principles of safety integration, with the consequence that information addressed to the operator cannot substitute for design.

The criterion compares times, not technologies. It does not age with definitions of autonomy, and it does not require amendment as architectures change. It applies identically to the millisecond mechanical hazard and to the self-evolving controller.

6.2 The one value that has never been consolidated

The comparison is exact in form and rests on inputs of unequal maturity. Two of the three are already standardised in detail; the third is not standardised at all, and the asymmetry is itself part of the finding.

The response time of a halting mechanism is quantified with considerable precision. ISO 13849-1 and the functional-safety family define safety functions, performance levels, architectural categories and response times; this is the most developed part of machine-safety engineering. The interval the hazard allows is likewise routinely calculated by whoever designs an installation: stopping performance, overtravel, separation distances — the material ISO 13855 exists to organise.

The window a human being requires has never been consolidated in the same way. The human-factors literature supplies orders of magnitude, and Section 2.2 relies on nothing finer, but no harmonised standard establishes minimum credible human response times as reference values for a defined operating configuration, in the way approach speeds and system response times are established. Machine safety has quantified what belongs to the machine and left unquantified what belongs to the person in front of it, while building the whole protective architecture on that person's capacity to act.

The comparison needs three numbers. Two of them can be looked up. Standards specify how fast a halting mechanism responds, and how to calculate the time a hazard leaves before its effect can no longer be prevented. The third cannot be looked up. No standard says how long a person needs to notice a hazard, decide, and act, in a given working position.

ISO 13855 shows that a number of this kind is not foreign to the field. It fixes how fast a body part is assumed to approach a hazard, and compares that with how fast the system responds. The missing number is of the same kind, and would be used in the same way.

Two conventions would have to be settled in the course of that work, and neither is decided here. The first is which value of the available interval governs, since a hazardous action has a nominal duration and a credible worst case which may differ by an order of magnitude. The second is what margin must separate the mechanism's response time from the closing of the interval: a mechanism that finishes marginally inside it has not arrested anything, since the tolerances of any real installation will erase the difference. Both are engineering determinations, and both are the ordinary business of a standards committee.

6.3 Third parties

Article 3(1) LSPro extends the product-safety duty to users and third parties. Section 1910.212(a)(1) protects the operator and other employees in the machine area. The protection of bystanders therefore needs no separate argument.

A halting mechanism satisfying 2.4 does not ask who is standing in the hazard zone. Protection of the person at the machine and of the persons around it is one design outcome seen from two positions, provided that the exposure of those others is included in the configuration for which the comparison was made. A protective stop dimensioned for the operating position has been shown to satisfy the comparison for that position and no other.

6.4 What remains open

If reliance on operator intervention must be justified by the comparison, the allocation of that justificatory burden remains open — in conformity assessment, and in litigation after an accident. Section 2.7 states where the burden lies as a matter of logic. Where it lies as a matter of procedure, in each of the three systems, is not settled here.

7. Conclusion

Machine safety law protects the operator against time. A stop function, an emergency device, a guard: each is a way of holding open an interval in which someone can still act.

Whether that interval is long enough is a question of fact. The hazard fixes what is available. Physiology fixes what is required. Where the first is shorter than the second, an architecture whose last layer is the operator's hand protects no one, however complete its documentation. The capacity to halt must then lie in a mechanism fast enough to act within the interval and separate enough from the source of the hazard for its speed to mean something. Where no such mechanism can exist, that configuration cannot be shown to satisfy the duty at all.

European, Swiss and American law each require this result. The international standards that carry machine safety into systems agreeing on little else contain the mechanisms it calls for, in technical detail. What is stated nowhere is the measurement. A duty discharged through an operation needs the time that operation takes, and a duty that cannot be connected to the facts of a particular machine fails silently, one formally compliant installation at a time.

Systemic arrestability creates nothing. It states the comparison and asks for it to be demonstrated. A machine that relies on human intervention in conditions where human intervention is physically excluded has not discharged the duty of safe design. It has only documented it. The operator whose hand can no longer reach the stop in time has not lost the right to a machine that stops. The law that promised the stop has only to be read as meaning it.

NOTES

- [1] Regulation (EU) 2023/1230, Art. 2(2)(l), excluding machinery or related products specially designed and constructed for military or police purposes; an equivalent exclusion appears in Directive 2006/42/EC, Art. 1(2)(g). The exclusion is jurisdictional, not logical: the comparison of Section 2 does not change its nature at the boundary of civil law, and a person operating or accompanying such a system stands in the same factual position of exposure wherever it is deployed — holding the controls confers no temporal advantage where the interval is closed. That is a statement of fact and not of obligation. Whether, and through which instruments, an arrestability criterion has purchase beyond civil product-safety law is a question this paper does not pursue; nothing here derives any such obligation from civil product-safety law.
- [2] The author has developed a parallel analysis of temporal compression in a different legal regime, where the protected capacity is the recognition of surrender rather than the halting of a machine: G. Nardacci, *Lawful Operational Safeguards in AI Systems*, Zenodo concept DOI 10.5281/zenodo.21932439. The two analyses are independent, rest on separate legal anchors, and share only a temporal structure by analogy. Nothing in the present paper derives any obligation for the conduct of hostilities from civil product-safety law, and nothing in it derives any civil product-safety obligation from the law of armed conflict. Further documentation: humanflag.org.
- [3] M. Green, "How Long Does It Take to Stop?" Methodological Analysis of Driver Perception-Brake Times", *Transportation Human Factors* 2(3) (2000) 195–216; H. Summala, "Brake Reaction Times and Driver Behavior Analysis", *ibid.*, 217–226, reporting on-road braking latencies of 1.0–1.3 s in fairly urgent situations. The chain structure is independently established: choice among alternatives lengthens response time logarithmically with the number of options (W.E. Hick, *Quarterly Journal of Experimental Psychology* 4 (1952) 11–26; R. Hyman, *Journal of Experimental Psychology* 45 (1953) 188–196), and the perceptual, cognitive and motor components have been separately measured, each on the order of tens to hundreds of milliseconds.

(S.K. Card, T.P. Moran, A. Newell, *The Psychology of Human-Computer Interaction*, 1983, ch. 2). That the literature yields orders of magnitude rather than fixed constants is the standard position of human-factors engineering: C.D. Wickens et al., *An Introduction to Human Factors Engineering*, 2nd ed., 2004.

- [4] European Commission, Machinery Working Group document MD WG 2010-03 EN rev. 2 (approved 5–6 November 2014), on emergency stop devices under Annex I, section 1.2.4.3 of Directive 2006/42/EC, quoting the *Guide to application of the Machinery Directive*.
- [5] Case T-337/13, *CSF Srl v European Commission*, judgment of the General Court (Third Chamber) of 15 July 2015, in particular the holdings summarised at paras 64–70 and 83–84.
- [6] Regulation (EU) 2023/1230, Annex III, section 1.2.1, third subparagraph, point (c); applicable from 20 January 2027 (Art. 54).
- [7] Loi fédérale sur la sécurité des produits du 12 juin 2009 (LSPro) / Legge federale sulla sicurezza dei prodotti, RS 930.11, in force since 1 July 2010.
- [8] 29 CFR § 1910.212 (General requirements for all machines), Subpart O — Machinery and Machine Guarding.
- [9] Akron Brick & Block Co., 3 BNA OSHC 1876, 1877–78 (No. 4859, 1976); Rockwell Int'l Corp., 9 BNA OSHC 1092, 1097–98 (No. 12470, 1980); Fabricated Metal Products, Inc., 18 BNA OSHC 1072, 1073–74 (No. 93-1853, 1997). For a recent application, see Aerospace Testing Alliance, OSHRC Docket No. 16-1167 (21 September 2020).
- [10] Technical Regulation of the Customs Union TR CU 010/2011, "On safety of machinery and equipment", adopted by Decision No. 823 of the Customs Union Commission of 18 October 2011, in force since 15 February 2013. Quotations follow published English translations of the official Russian text; the original is authoritative.
- [11] GB/T 15706 (ISO 12100, IDT); GB/T 5226.1 (IEC 60204-1, IDT); GB/T 16855.1 (ISO 13849-1, IDT), People's Republic of China national adoptions.
- [12] Work Safety Ordinance (New Version), 5730-1970 (Israel); mandatory Israeli standards declared under the Standards Law are adoptions of the corresponding IEC/ISO standards by the Standards Institution of Israel. Cited at the level of architecture only; primary-source verification in Hebrew remains outstanding.
- [13] The allocation of the safety duty to construction rather than to conduct long predates the frameworks that codify it. Deuteronomy 22:8 commands the builder of a new house to construct a parapet on its roof — an obligation attaching to the act of building, before any negligence in use can occur. Within the legal heritage to which Israel's Foundations of Law Act, 5740-1980, assigns a residual interpretive role, that duty remains recognised as interpretatively relevant. The reference is offered as a historical illustration of a design-based allocation of safety duties, not as a source of contemporary machine-safety obligations.

All normative citations have been verified against the official texts (EUR-Lex consolidated text CELEX 02023R1230; ecf.gov; fedlex.admin.ch / admin.ch) as of July 2026.

REFERENCES

Legislation and regulations. Directive 2006/42/EC, OJ L 157, 9.6.2006, p. 24; Regulation (EU) 2023/1230, OJ L 165, 29.6.2023; LSPro, RS 930.11; 29 CFR § 1910.212, Subpart O; TR CU 010/2011 (EAEU); GB/T 15706, GB/T 5226.1, GB/T 16855.1 (PRC national adoptions); Work Safety Ordinance (New Version), 5730-1970 (Israel).

Standards. ISO 12100; ISO 13849-1; ISO 13855 (current edition adopted in Switzerland as SN EN ISO 13855:2025); IEC 60204-1.

Case law. Case T-337/13, *CSF Srl v European Commission*; Akron Brick & Block Co.; Rockwell International Corp.; Fabricated Metal Products, Inc.; Aerospace Testing Alliance.

Official guidance. European Commission, *Guide to application of the Machinery Directive 2006/42/EC*, 2nd ed., June 2010; Machinery Working Group document MD WG 2010-03 EN rev. 2.

Literature. Card, Moran & Newell (1983); Green (2000); Hick (1952); Hyman (1953); Summala (2000); Wickens et al. (2004).

REVISION NOTE — VERSION 7, SEPTEMBER 2026

This version restates the argument in causal sequence. The comparison is derived first, without reference to any legal system; the European, Swiss and United States frameworks are then shown to impose its result. One interval and the windows that must fit inside it replace the earlier terminology. Independence is added as the condition under which a mechanism's response time is measurable at all. Where no admissible mechanism exists, the consequence is stated as a demonstration that cannot be produced, rather than left as an open question. The three negative clarifications of the earlier version are removed, as each follows from the steps.

An illustrative example drawn from a close-in defensive system has been removed: the interval it described was that of the engagement sequence, not the interval this paper measures.

Supersedes Version 6.